

Supplementary Information

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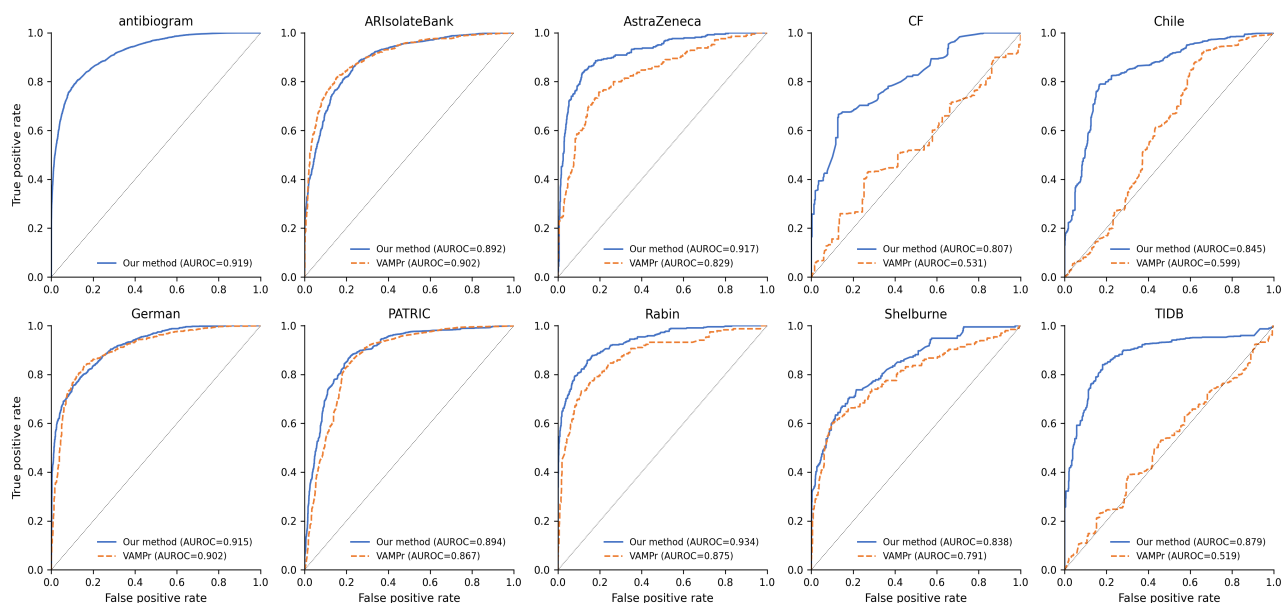
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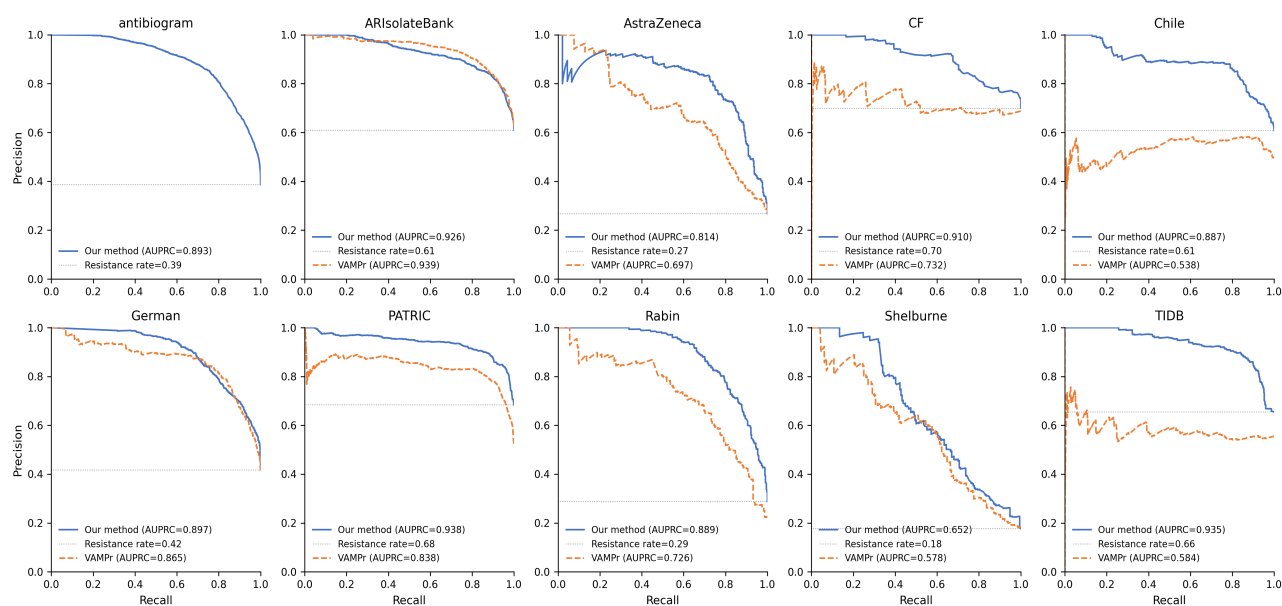
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Supplementary figures



SFig1. Per-cohort receiver-operating-characteristic (ROC) curves for different methods. Each panel shows one cohort of the cohort-holdout test set; curves compare our method (drug-conditioned cross-attention) and VAMPr. AUROC, area under the receiver-operating-characteristic curve. VAMPr is omitted from the antibiogram panel because antibiogram is the published VAMPr training cohort.



SFig2. Per-cohort precision-recall (PR) curves for different methods. Each panel shows one cohort; curves compare our method and VAMPr. Dotted horizontal line marks the cohort resistance rate (= positive-class fraction), the baseline AUPRC of a random classifier. AUPRC, area under the precision-recall curve.

Supplementary tables

STable1. Cross-cohort performance comparison across different methods. Per-cohort metrics for our method and VAMPr on the 10 cohort-holdout test sets. Twenty rows = 10 cohorts × 2 methods. VAMPr is omitted from the antibiogram cohort (its published training set), and the corresponding cells are marked with an em-dash. AUROC is reported with a 95% bootstrap confidence interval (CI). MCC, Matthews correlation coefficient.

cohort	method	accuracy	balanced accuracy	AUROC	AUROC CI lower 95	AUROC CI upper 95	AUPRC	F1	MCC	sensitivity	specificity	resistance rate
antibiogram	Our method	0.8610	0.8335	0.9192	0.9144	0.9240	0.8926	0.7972	0.6870	0.7463	0.9207	0.3866
antibiogram	VAMPr	—	—	—	—	—	—	—	—	—	—	—
ARIsolateBank	Our method	0.8230	0.8129	0.8917	0.8760	0.9072	0.9259	0.8595	0.6321	0.8735	0.7524	0.6091
ARIsolateBank	VAMPr	0.8290	0.8232	0.9024	0.8852	0.9195	0.9389	0.8612	0.6380	0.8434	0.8030	0.6308
AstraZeneca	Our method	0.8730	0.8397	0.9169	0.9007	0.9325	0.8136	0.7639	0.6767	0.7688	0.9106	0.2679
AstraZeneca	VAMPr	0.8090	0.7762	0.8286	0.7954	0.8600	0.6967	0.6682	0.5356	0.7048	0.8477	0.2734
CF	Our method	0.7240	0.7360	0.8073	0.7629	0.8477	0.9099	0.7800	0.4362	0.7030	0.7690	0.6990
CF	VAMPr	0.5090	0.5712	0.5306	0.4805	0.5820	0.7316	0.5322	0.1373	0.4061	0.7362	0.6876
Chile	Our method	0.7900	0.7978	0.8455	0.8189	0.8702	0.8867	0.8365	0.5916	0.8256	0.7700	0.6081
Chile	VAMPr	0.6030	0.5995	0.5986	0.5659	0.6291	0.5381	0.6539	0.2106	0.7643	0.4346	0.4964
German	Our method	0.8300	0.8217	0.9147	0.9016	0.9269	0.8989	0.7913	0.6481	0.7728	0.8705	0.4178
German	VAMPr	0.8010	0.8131	0.9018	0.8824	0.9182	0.8646	0.7857	0.6170	0.8806	0.7457	0.4133
PATRIC	Our method	0.8290	0.7846	0.8936	0.8882	0.8986	0.9398	0.8968	0.6376	0.9588	0.6105	0.6838
PATRIC	VAMPr	0.7990	0.7913	0.8673	0.8615	0.8732	0.8379	0.8293	0.6117	0.9303	0.6523	0.5261
Rabin	Our method	0.8550	0.8541	0.9336	0.9140	0.9508	0.8888	0.7731	0.6751	0.8479	0.8602	0.2880
Rabin	VAMPr	0.7930	0.8000	0.8747	0.8420	0.9052	0.7262	0.6335	0.5231	0.8128	0.7873	0.2202
Shelburne	Our method	0.8240	0.7593	0.8378	0.7949	0.8780	0.6522	0.5670	0.4647	0.6649	0.8537	0.1770
Shelburne	VAMPr	0.7000	0.7152	0.7906	0.7410	0.8397	0.5779	0.4731	0.3410	0.7398	0.6906	0.1823
TIDB	Our method	0.8290	0.8195	0.8785	0.8567	0.8988	0.9354	0.8694	0.6315	0.8573	0.7817	0.6553
TIDB	VAMPr	0.5190	0.5363	0.5186	0.4847	0.5530	0.5845	0.4631	0.0760	0.3745	0.6980	0.5541

STable2. Cohort × species summary. Counts of isolates, observations, drugs, and resistant observations, together with the resistance rate, for each (cohort, species) pair. 32 rows in total.

cohort	species	n isolates	n observations	n drugs	n resistant	resistance rate
Antibiogram	Acinetobacter baumannii	653	2446	8	919	0.3757
Antibiogram	Enterobacter cloacae	79	618	10	288	0.4660
Antibiogram	Escherichia coli	350	4329	20	2229	0.5149
Antibiogram	Klebsiella aerogenes	68	173	3	65	0.3757
Antibiogram	Klebsiella pneumoniae	344	3810	14	2971	0.7798
Antibiogram	Pseudomonas aeruginosa	83	685	9	436	0.6365
Antibiogram	Salmonella enterica	1349	10370	8	2182	0.2104
Antibiogram	Staphylococcus aureus	31	86	3	33	0.3837
Antibiogram	Streptococcus pneumoniae	317	1470	6	150	0.1020
ARIsolateBank	Acinetobacter baumannii	3	41	14	34	0.8293
ARIsolateBank	Citrobacter freundii	2	16	11	5	0.3125
ARIsolateBank	Enterobacter cloacae	18	256	17	164	0.6406
ARIsolateBank	Escherichia coli	56	875	20	470	0.5371
ARIsolateBank	Klebsiella oxytoca	2	13	9	8	0.6154
ARIsolateBank	Klebsiella pneumoniae	75	1242	21	833	0.6707
ARIsolateBank	Pseudomonas aeruginosa	46	516	12	346	0.6705
ARIsolateBank	Staphylococcus aureus	97	791	10	424	0.5360
AstraZeneca	Pseudomonas aeruginosa	384	1437	4	385	0.2679
CF	Pseudomonas aeruginosa	116	1050	11	734	0.6990
Chile	Pseudomonas aeruginosa	281	2008	8	1221	0.6081
German	Pseudomonas aeruginosa	626	3329	13	1391	0.4178
PATRIC	Klebsiella pneumoniae	1666	23739	16	16232	0.6838
Rabin	Acinetobacter baumannii	23	144	7	110	0.7639
Rabin	Enterobacter cloacae	3	34	12	4	0.1176
Rabin	Escherichia coli	67	740	13	163	0.2203
Rabin	Klebsiella pneumoniae	27	291	12	83	0.2852
Rabin	Pseudomonas aeruginosa	48	298	7	74	0.2483

cohort	species	n isolates	n observations	n drugs	n resistant	resistance rate
Shelburne	Enterobacter cloacae	13	39	3	10	0.2564
Shelburne	Escherichia coli	31	93	3	35	0.3763
Shelburne	Klebsiella pneumoniae	24	72	3	36	0.5000
Shelburne	Pseudomonas aeruginosa	129	892	9	113	0.1267
TIDB	Pseudomonas aeruginosa	205	1967	11	1289	0.6553

STable4. Per-drug performance on the cohort-holdout evaluation. Per-drug accuracy, AUROC, F1, MCC, sensitivity, specificity and resistance rate for the 36 training drugs (one row per drug). Metrics whose value is undefined for a drug — typically because the drug has a single resistance class in the evaluation subset (e.g. polymyxin B, chloramphenicol) — are rendered as em-dash.

drug	resistance rate	accuracy	AUROC	F1	MCC	sensitivity	specificity
polymyxin B	0.0000	1.0000	—	0.0000	0.0000	—	1.0000
ampicillin	0.9882	0.9876	0.9689	0.9937	0.4172	0.9949	0.3810
colistin	0.0169	0.9831	0.6397	0.0000	0.0000	0.0000	1.0000
amoxicillin	0.0234	0.9766	0.5000	0.0000	0.0000	0.0000	1.0000
doxycycline	0.0400	0.9600	0.8749	0.0000	0.0000	0.0000	1.0000
cefazolin	0.9532	0.9508	0.9201	0.9747	0.1155	0.9950	0.0513
ceftriaxone	0.9237	0.9379	0.9368	0.9671	0.4516	0.9882	0.3286
nitrofurantoin	0.9271	0.9233	0.5154	0.9601	-0.0174	0.9959	0.0000
gentamicin	0.4663	0.9114	0.7359	0.8995	0.8250	0.8508	0.9643
cefuroxime	0.0955	0.9045	0.6107	0.0000	0.0000	0.0000	1.0000
kanamycin	0.0963	0.9037	0.7568	0.0000	0.0000	0.0000	1.0000
levofloxacin	0.8505	0.8963	0.7917	0.9384	0.6130	0.9285	0.7135
ertapenem	0.3128	0.8930	0.7182	0.8434	0.7693	0.9211	0.8802
amikacin	0.1872	0.8878	0.7666	0.6244	0.5896	0.4982	0.9776
ciprofloxacin	0.8168	0.8854	0.8659	0.9291	0.6323	0.9189	0.7363
tigecycline	0.1354	0.8750	0.4295	0.1429	0.2592	0.0769	1.0000
rifampin	0.1392	0.8608	0.7092	0.0000	0.0000	0.0000	1.0000
ceftazidime	0.7105	0.8520	0.7676	0.8979	0.6311	0.9157	0.6956
tobramycin	0.4678	0.8211	0.7531	0.8242	0.6532	0.8962	0.7551
aztreonam	0.7488	0.8126	0.7183	0.8797	0.4655	0.9154	0.5062
doripenem	0.5824	0.8123	0.7970	0.8464	0.6108	0.8882	0.7064
ceftiofur	0.1956	0.8044	0.7905	0.0000	0.0000	0.0000	1.0000
penicillin	0.9588	0.7732	0.6142	0.8721	-0.0990	0.8065	0.0000
cefotaxime	0.6816	0.7592	0.9119	0.8218	0.4510	0.8144	0.6410
cefepime	0.6322	0.7359	0.7459	0.8119	0.4053	0.9012	0.4517
clindamycin	0.3780	0.7195	0.8292	0.7013	0.4870	0.8710	0.6275
piperacillin	0.5086	0.7184	0.7926	0.7118	0.4388	0.6836	0.7544
meropenem	0.4169	0.7109	0.6806	0.7049	0.4525	0.8282	0.6271
minocycline	0.2800	0.7000	0.6483	0.5161	0.3047	0.5714	0.7500
ofloxacin	0.2234	0.6915	0.6651	0.5915	0.5031	1.0000	0.6027
imipenem	0.4064	0.6733	0.4226	0.6752	0.3986	0.8354	0.5624
tetracycline	0.5200	0.6319	0.6199	0.6870	0.2648	0.7769	0.4748
erythromycin	0.3692	0.6308	0.8026	0.0000	0.0000	0.0000	1.0000
chloramphenicol	0.0000	0.6000	0.9257	0.0000	0.0000	—	0.6000
oxacillin	0.5361	0.5773	0.8036	0.4810	0.2087	0.3654	0.8222
cefoxitin	0.5513	0.5611	0.8098	0.7064	0.0680	0.9577	0.0739

STable8. Model and baseline architectures, optimisers and hardware. Concrete training settings for our method (FARM) and each baseline. Nine rows total. Source files and full hyperparameter values are listed in the build scripts.

method	category	drug input	gene input	architecture	optimizer	learning rate	hardware
FARM (ours)	Deep learning — graph + cross-attention	SMILES → molecular graph (atoms, bond types, pair-wise shortest-path distance); max 85 atoms; 16-symbol atom vocabulary	1,962 KO-cluster features per isolate (name + presence + mutation one-hots)	2× Drug-Graphformer encoder blocks (num_hiddens=60, fused atom/edge/relative-position attention; mixing weights $\alpha/\beta/\delta = 0.6/0.2/0.3$ for atom/edge/rel) → 2× drug-gene cross-attention decoder blocks (num_hiddens=30) → sigmoid head; self-attention temperature T=9; saliency self-score blend $\lambda=0.5$	Adam	5e-5	NVIDIA A100-SXM4-80GB × N (TF MirroredStrategy)
MLP	Deep learning — dense	Atom one-hot flattened (max 114 atoms × 16 vocab)	Gene presence/mutation one-hots flattened (3,287 dim)	Dense(1024)-ReLU → Dropout(0.5) → Dense(512)-ReLU → Dropout(0.5) → Dense(256)-ReLU → Dropout(0.5) → Dense(1)-sigmoid	Adam	1e-5	NVIDIA A100-SXM4-80GB
CNN	Deep learning — 1D convolution	Atom one-hot sequence (max 114 × 16)	Gene one-hots flattened → Dense(256)-ReLU	SMILES: Conv1D(filters=128, kernel=5, padding=same) → GlobalMaxPool1D; concat with gene branch → Dense(256)-ReLU → Dense(1)-sigmoid	Adam	1e-5	NVIDIA A100-SXM4-80GB
Transformer (baseline)	Deep learning — sequence transformer	Atom one-hot sequence (max 114 × 16)	Gene one-hots flattened → Dense(64)-ReLU	Dense(64)→positional encoding → 2× (MultiHeadAttention(heads=4, key_dim=64) + LN + Dense(64)-ReLU + LN) → GlobalMaxPool1D; concat with gene branch → Dense(128)-ReLU → Dense(1)-sigmoid	Adam	1e-5	NVIDIA A100-SXM4-80GB
Random Forest	Classical ML — ensemble	Atom one-hot flattened (114 × 16)	Gene presence/mutation one-hots flattened	sklearn RandomForestClassifier (n_estimators=200, default depth); concatenated drug + gene features	—	—	CPU (n_jobs=-1)
Elastic Net	Classical ML — linear	Atom one-hot flattened (114 × 16)	Gene presence/mutation one-hots flattened	sklearn LogisticRegression with elastic-net penalty (C=1.0, l1_ratio=0.5, saga solver)	SAGA	—	CPU (n_jobs=-1)
VAMPr	Specialized AMR — per-drug logistic regression	Drug identity (per-drug model; no SMILES)	Same 1,962 KO-cluster features as FARM (matched protocol)	Per-cohort × per-drug logistic regression with L1 (Kim et al. 2020 pipeline; one model per drug)	LIBLINEAR	—	CPU
ResFinder	Specialized AMR — rule/database	Drug identity (rule lookup; no SMILES)	BLAST against curated resistance-gene database	Sequence-similarity hit → AMR call (no trainable parameters)	—	—	CPU (BLAST)
RGI (CARD)	Specialized AMR — rule/database	Drug class lookup (no SMILES)	CARD homolog/protein-variant detection on assembled genome	Rule-based detection using CARD model types (no trainable parameters)	—	—	CPU

STable9. **PDB binding-atom validation of model self-attention.** One-sided Mann–Whitney U test comparing context-adjusted atom-level attention scores between PDB-derived binding atoms and non-binding atoms. Only drugs retained at $p < 0.05$ are listed (six drugs). The context-adjusted saliency follows Eq. 14 of Methods with $\lambda = 0.5$; attention logits are temperature-scaled at $T = 9$ before constructing the saliency matrix M .

drug	n atoms	U statistic	p value
chloramphenicol	20	57.0000	0.0102
doxycycline	32	175.5000	0.0161
tetracycline	32	144.5000	0.0183
aztreonam	28	112.5000	0.0205
ampicillin	24	104.5000	0.0298
ofloxacin	26	115.0000	0.0483

STable10. Training-drug metadata and resistance statistics. For each of the 36 training drugs: drug class (Figure 2A taxonomy), canonical SMILES, observation count, resistant / susceptible counts, and resistance rate across all 10 cohorts.

drug	drug class	SMILES	n observations	n resistant	n susceptible	resistance rate
amikacin	Aminoglycoside	C1C(C(C(C(C1NC(=O)C(CCN)O)OC2C(C(C(C(O2)CO)O)N)O)OC3C(C(C(C(O3)CN)O)O)O)N	4035	700	3335	0.1735
amoxicillin	Penicillin	CC1(C(N2C(S1)C(C2=O)NC(=O)C(C3=CC=C(C(=C3)O)N)C(=O)O)C	171	4	167	0.0234
ampicillin	Penicillin	CC1(C(N2C(S1)C(C2=O)NC(=O)C(C3=CC=CC=C3)N)C(=O)O)C	3471	2521	950	0.7263
aztreonam	Monobactam	CC1C(C(=O)N1S(=O)(=O)O)NC(=O)C(=NOC(C)(C)C(=O)O)C2=CSC(=N2)N	2796	2090	706	0.7475
cefazolin	Cephalosporin	CC1=NN=C(S1)SCC2=C(N3C(C(C3=O)NC(=O)CN4C=NN=N4)SC2)C(=O)O	1788	1697	91	0.9491
cefepime	Cephalosporin	C[N+](CCCC1)CC2=C(N3C(C(C3=O)NC(=O)C(=NOC)C4=CSC(=N4)N)SC2)C(=O)[O-]	3128	1998	1130	0.6387
cefotaxime	Cephalosporin	CC(=O)OCC1=C(N2C(C(C2=O)NC(=O)C(=NOC)C3=CSC(=N3)N)SC1)C(=O)O	617	506	111	0.8201
cefoxitin	Cephalosporin	COC1(C2N(C1=O)C(=C(CS2)COC(=O)N)C(=O)O)NC(=O)CC3=CC=CS3	3347	1353	1994	0.4042
ceftazidime	Cephalosporin	CC(C)(C(=O)O)ON=C(C1=CSC(=N1)N)C(=O)NC2C3N(C2=O)C(=C(C(S3)C[N+](4)=CC=CC=C4)C(=O)[O-])	3888	2847	1041	0.7323
ceftiofur	Cephalosporin	CON=C(C1=CSC(=N1)N)C(=O)NC2C3N(C2=O)C(=C(CS3)CSC(=O)C4=CC=CO4)C(=O)O	1416	277	1139	0.1956
ceftriaxone	Cephalosporin	CN1C(=NC(=O)C(=O)N1)SCC2=C(N3C(C(C3=O)NC(=O)C(=NOC)C4=CSC(=N4)N)SC2)C(=O)O	3526	2223	1303	0.6305
cefuroxime	Cephalosporin	CON=C(C1=CC=CO1)C(=O)NC2C3N(C2=O)C(=C(CS3)COC(=O)N)C(=O)O	178	17	161	0.0955
chloramphenicol	Amphenicol	C1=CC(=CC=C1C(C(CO)NC(=O)C(Cl)Cl)O)[N+](=O)[O-]	1418	99	1319	0.0698
ciprofloxacin	Fluoroquinolone	C1CC1N2C=C(C(=O)C3=CC(=C(C=C32)N4CCNCC4)F)C(=O)O	3966	3135	831	0.7905
clindamycin	Lincosamide	CCCC1CC(N(C1)C)C(=O)NC(C2C(C(C(C(O2)SC)O)O)O)C(C)Cl	422	51	371	0.1209
colistin	Polypeptide	CCC(C)CCCC(=O)NC(CCN)C(=O)NC(C(C)O)C(=O)NC(CCN)C(=O)N C1CCNC(=O)C(NC(=O)C(NC(=O)C(NC(=O)C(NC(=O)C(NC(=O)C(N C1=O)CCN)CC(C)C)CC(C)C)CCN)CCN)C(C)O	533	9	524	0.0169
doripenem	Carbapenem	CC1C2C(C(=O)N2C(=C1SC3CC(NC3)CNS(=O)(=O)N)C(=O)O)C(C)O	873	476	397	0.5452
doxycycline	Tetracycline	CC1C2C(C3C(C(=O)C(=C(C3(C(=O)C2=C(C4=C1C=CC=C4)O)O)O)C(=O)N)N(C)C)O	25	1	24	0.0400
ertapenem	Carbapenem	CC1C2C(C(=O)N2C(=C1SC3CC(NC3)C(=O)NC4=CC=CC(=C4)C(=O)O)C(=O)O)C(C)O	877	444	433	0.5063
erythromycin	Macrolide	CCC1C(C(C(C(=O)C(CC(C(C(C(C(=O)O1)C)OC2CC(C(C(O2)C)O)(C)OC)C)OC3C(C(CC(O3)C)N(C)C)O)(C)O)C)O)(C)O	409	151	258	0.3692
gentamicin	Aminoglycoside	CC(C1CCC(C(O1)OC2C(CC(C(C2O)OC3C(C(C(CO3)(C)O)NC)O)N)N)INC	4792	1698	3094	0.3543

drug	drug class	SMILES	n observations	n resistant	n susceptible	resistance rate
imipenem	Carbapenem	<chem>CC(C1C2CC(=C(N2C1=O)C(=O)O)SCCN=CN)O</chem>	3367	1517	1850	0.4505
kanamycin	Aminoglycoside	<chem>C1C(C(C(C(C1N)OC2C(C(C(C(O2)CN)O)O)O)OC3C(C(C(C(O3)C)O)N)O)N</chem>	1111	107	1004	0.0963
levofloxacin	Fluoroquinolone	<chem>CC1COC2=C3N1C=C(C(=O)C3=CC(=C2N4CCN(CC4)C)F)C(=O)O</chem>	3329	2688	641	0.8074
meropenem	Carbapenem	<chem>CC1C2C(C(=O)N2C(=C1SC3CC(NC3)C(=O)N(C)C)C(=O)O)C(C)O</chem>	4823	2079	2744	0.4311
minocycline	Tetracycline	<chem>CN(C)C1C2CC3CC4=C(C=CC(=C4C(=C3C(=O)C2(C(=C(C1=O)C(=O)N)O)O)O)O)N(C)C</chem>	100	28	72	0.2800
nitrofurantoin	Nitrofuran	<chem>C1C(=O)NC(=O)N1N=CC2=CC=C(O2)[N+](=O)[O-]</chem>	782	725	57	0.9271
ofloxacin	Fluoroquinolone	<chem>CC1COC2=C3N1C=C(C(=O)C3=CC(=C2N4CCN(CC4)C)F)C(=O)O</chem>	94	21	73	0.2234
oxacillin	Penicillin	<chem>CC1=C(C(=NO1)C2=CC=CC=C2)C(=O)NC3C4N(C3=O)C(C(S4)(C)C)C(=O)O</chem>	97	52	45	0.5361
penicillin	Penicillin	<chem>CC1(C(N2C(S1)C(C2=O)NC(=O)CC3=CC=CC=C3)C(=O)O)C</chem>	97	93	4	0.9588
piperacillin	Penicillin	<chem>CCN1CCN(C(=O)C1=O)C(=O)NC(C2=CC=CC=C2)C(=O)NC3C4N(C3=O)C(C(S4)(C)C)C(=O)O</chem>	348	177	171	0.5086
polymyxin B	Polypeptide	<chem>CCC(C)CCCC(=O)NC(CCN)C(=O)NC(C(C)O)C(=O)NC(CCN)C(=O)NC1CCNC(=O)C(NC(=O)C(NC(=O)C(NC(=O)C(NC(=O)C(NC(=O)C(NC(=O)C(NC1=O)CCN)CC2=CC=CC=C2)CC(C)C)CCN)CCN)C(C)O</chem>	81	0	81	0.0000
rifampin	Rifamycin	<chem>CC1C=CC=C(C(=O)NC2=C(C(=C3C(=C2O)C(=C(C4=C3C(=O)C(O4)(OC=CC(C(C(C(C(C1O)C)O)C)OC(=O)C)C)OC)C)O)C=NN5C(CN(CC5)C)C</chem>	79	11	68	0.1392
tetracycline	Tetracycline	<chem>CC1(C2CC3C(C(=O)C(=C(C3(C(=O)C2=C(C4=C1C=CC=C4O)O)O)C(=O)N)N(C)C)O</chem>	4230	2020	2210	0.4775
tigecycline	Tetracycline	<chem>CC(C)(C)NCC(=O)NC1=CC(=C2CC3CC4C(C(=O)C(=C(C4(C(=O)C3=C(C2=C1O)O)O)O)C(=O)N)N(C)C)N(C)C</chem>	96	13	83	0.1354
tobramycin	Aminoglycoside	<chem>C1C(C(C(C(C1N)OC2C(C(C(C(O2)CO)O)N)O)O)OC3C(CC(C(O3)CN)O)N)N</chem>	3560	1609	1951	0.4520